(i) When $f(x) = 1$, $g(x) = e^x$, $a = 0$, $b = t$:

$$\left(\int_0^t e^x\,dx\right)^2 \le \left(\int_0^t 1\,dx\right)\left(\int_0^t e^{2x}\,dx\right)$$

$$\implies (e^t - 1)^2 \le t \cdot \frac{e^{2t} - 1}{2}$$

$$\implies (e^t - 1)^2 \le \frac{t}{2}(e^t - 1)(e^t + 1)$$

$\because t > 0 \quad \therefore e^t - 1 > 0$

$$\frac{e^t - 1}{e^t + 1} \le \frac{t}{2}$$

(ii) When $f(x) = x$, $g(x) = e^{-\frac{x^2}{4}}$, $a = 0$, $b = 1$:

$$\left(\int_0^1 xe^{-\frac{x^2}{4}}\,dx\right)^2 \le \left(\int_0^1 x^2\,dx\right)\left(\int_0^1 e^{-\frac{x^2}{2}}\,dx\right)$$

$$\because \int_0^1 x^2\,dx = \frac{1}{3} \quad \&\& \quad \int_0^1 xe^{-x^2/4}\,dx = \left[-2e^{-x^2/4}\right]_0^1 = 2(1 - e^{-1/4})$$

$$\therefore 4(1 - e^{-1/4})^2 \le \frac{1}{3}\int_0^1 e^{-\frac{x^2}{2}}\,dx$$

$$\implies \int_0^1 e^{-x^2/2}\,dx \ge 12(1 - e^{-1/4})^2$$

(iii) When $f(x) = 1$, $g(x) = \sqrt{\sin x}$, $a = 0$, $b = \frac{\pi}{2}$:

$$\left(\int_0^{\pi/2} \sqrt{\sin x}\,dx\right)^2 \le \left(\int_0^{\pi/2} 1\,dx\right)\left(\int_0^{\pi/2} \sin x\,dx\right)$$

$$\implies \left(\int_0^{\pi/2} \sqrt{\sin x}\,dx\right)^2 \le \frac{\pi}{2}$$

$$\because \sin x > 0, \quad x \in [0, \frac{\pi}{2}]$$

$$\therefore \int_0^{\pi/2} \sqrt{\sin x}\,dx \le \sqrt{\frac{\pi}{2}}$$

When $f(x) = \cos x$, $g(x) = (\sin x)^{\frac{1}{4}}$, $a = 0$, $b = \frac{\pi}{2}$:

$$\left(\int_0^{\pi/2} (\sin x)^{1/4}\cos x\,dx\right)^2 \le \left(\int_0^{\pi/2} \cos^2 x\,dx\right)\left(\int_0^{\pi/2} \sqrt{\sin x}\,dx\right)$$

$$\because \int_0^{\pi/2} (\sin x)^{1/4} \cos x \, dx = \left[\frac{4}{5} (\sin x)^{5/4} \right]_0^{\pi/2} = \frac{4}{5}$$

$$\because \int_0^{\pi/2} \cos^2 x \, dx = \int_0^{\pi/2} \frac{1 + \cos(2x)}{2} \, dx = \left[\frac{x}{2} + \frac{\sin(2x)}{4} \right]_0^{\pi/2} = \frac{\pi}{4}$$

$$\therefore \left(\frac{4}{5} \right)^2 \leq \frac{\pi}{4} \cdot \int_0^{\pi/2} \sqrt{\sin x} \, dx$$

$$\implies \frac{16}{25} \leq \frac{\pi}{4} \cdot \int_0^{\pi/2} \sqrt{\sin x} \, dx$$

$$\implies \frac{64}{25\pi} \leq \int_0^{\pi/2} \sqrt{\sin x} \, dx$$

$$\implies \frac{64}{25\pi} \leq \int_0^{\pi/2} \sqrt{\sin x} \, dx \leq \sqrt{\frac{\pi}{2}}$$

This solution was submitted by **Jon**.

Published and refined by **Jon**.

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